Original article

Predictors of outcome in the surgical treatment for epilepsy

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Keywords: *epilepsy surgery; seizure outcome; focal cortical dysplasia; magnetic resonance imaging; electroencephalography*

Background Knowledge about factors influencing the prognosis of resective epilepsy surgery can be used to identify which patients are most suitable for surgical treatment. The aim of this study was to identify preoperative prognostic factors associated with the chance of achieving long-term seizure freedom.

Methods We retrospectively reviewed seizure outcomes and clinical, electroencephalography (EEG), magnetic resonance imaging (MRI), histopathology, and surgical variables from 99 epilepsy surgery patients with at least one year of postoperative follow-up. Seizure outcomes were categorized based on the modified classification by the International League Against Epilepsy.

Results We found that the seizure-free rate was 27.9% after one year, and that it stabilized at about 20.0% between two and six years after surgery. Univariate analysis showed that medial temporal lobe epilepsy with hippocampal sclerosis, MRI with visible focal lesions concordant with EEG, and regional ictal EEG and electrocorticography patterns were associated with a favorable surgical outcome. On the other hand, seizure recurrence within six months, incomplete focus resection, and surgical complications were associated with a poor outcome. Multivariate analysis showed that medial temporal lobe epilepsy with hippocampal sclerosis and MRI with visible focal lesions were independent presurgical predictors of a favorable outcome (P <0.01). Seizure recurrence within six months was the only significant independent predictor associated with a poor outcome (P <0.01).

Conclusion Hippocampal sclerosis and abnormal MRI findings are strongly associated with a favorable surgical outcome, whereas seizure recurrence within six months is associated with a poor outcome.

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The last two decades have witnessed remarkable advances in the evaluation and treatment of patients with refractory epilepsy.[1] Surgical treatment is an option in a subgroup of patients with medically refractory epilepsy.[2] The appropriate selection of patients likely to be helped by surgery is crucial to best help epilepsy patients. Selection of surgery candidates is based on the identification of predictors of favorable or unfavorable postoperative outcomes. Significant predictors of favorable postsurgical outcomes identified in previous studies include early diagnosis of intractable epilepsy,[3] regional ictal electroencephalography (EEG) patterns,[4] and temporal lobe epilepsy with hippocampal sclerosis (HS).[5] To further address these issues, we analyzed histopathological and clinical data, EEG, magnetic resonance imaging (MRI), and surgical variables in relation to seizure outcome, in a cohort of 99 epilepsy patients with at least one year of postoperative follow-up.

METHODS

Patient selection

Patients who underwent resective epilepsy surgery at Shanghai Renji Hospital, Shanghai, China from January 2003 to December 2008 were retrospectively reviewed. Follow-up of at least one year was required. Data collected included gender, age at initial seizure onset, age at the time of surgery, seizure duration, the presence of auras and of secondary seizure generalization, family history of epilepsy, seizure semiology and frequency, initial precipitating incident (IPI), preoperative neurologic signs, side of surgery, date of seizure recurrence, date of last follow-up, results of pre-operative EEG and MRI, and postoperative pathology. The occurrence of acute post-operative seizures (APO), defined as seizures occurring within the first post-operative week, was also recorded.[6] Medical intractability was defined as 18 months of unsuccessful treatment with more than two leading antiepileptic drugs at adequate doses, and a seizure frequency of more than one per month.[7] IPIs were defined as events involving loss of consciousness for more than 30 minutes or a change in cognitive function for more than 4 hours.[8]

Pre-operative patient evaluation protocol

A complete history and neurological examination were obtained in every patient. All patients underwent

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preoperative scalp video EEG using the standard 10–20 system of electrodes with additional electrodes applied in selected cases. Interictal epileptiform discharges and ictal EEG patterns were classified as either regional (appearing exclusively over a single lobe or in two contiguous regions, such as centroparietal discharges) or non-regional (e.g. multilobar, hemispheric, or generalized). Patients with multiple ictal EEG patterns (such as regional and non-unilateral patterns) were always considered non-regional. Intraoperative electrocorticography (ECoG) was performed in each case to either confirm removal of epileptogenic cortex, or to avoid eloquent cortex resection. The occurrence of multifocal seizures during invasive evaluation was recorded.

Magnetic resonance imaging

All patients underwent brain MRI. Standard MRI was performed on either a 1.0-T or a 3.0-T unit with conventional spin-echo T1-weighted sagittal and T2-weighted axial and coronal sequences and fluid-attenuated inversion recovery (FLAIR) sequences.

Surgical and pathological subgroups

All patients underwent resective surgery. The surgical area was determined based on the clinical, neuroimaging, and electrophysiological results. In our study, the resection margin for epilepsy was defined by: (1) the presence of either a discrete lesion on MRI with compatible ictal EEG, or a massive and exclusive ictal-onset zone confirmed by intracranial EEG, and (2) the absence of eloquent cortex. Complete resection was defined by removal of the MRI abnormality as well as the electrophysiological abnormalities. In patients with normal MRI, complete resection of brain tissue involved in ictal onset and most frequent interictal abnormalities was required.[9] All paraffin-embedded specimens were reexamined by a senior neuropathologist at Shanghai Renji Hospital.

Outcome definition

Follow-up information was obtained from clinic visits and patient phone calls. The typical follow-up schedule consisted of a clinic visit at 6 and 12 months post-operatively in the first year, and then annually in the ensuing years in all patients. The primary outcome was the time to first seizure recurrence. In patients who underwent multiple operations, the outcome of the latest operation was used as the final outcome. Surgical outcomes were classified according to the International League Against Epilepsy (ILAE):[10] Class 1a: completely seizure-free since surgery and no auras; Class 1: completely seizure-free, excluding early post-operative seizures; Class 2: only auras and no other seizures; Class 3: 1–3 seizure days per year, with or without auras; Class 4: four seizure days per year to 50% reduction in baseline seizure days, with or without auras; Class 5: <50% reduction in baseline seizure days to 100% increase in baseline seizure days, with or without auras; Class 6: >100% increase in baseline seizure days, with or without auras.

Statistical analysis

Data were analyzed in SPSS 16.0 (SPSS Inc., USA) and then reported as mean ± standard deviation (SD), frequency and percentile for description. Kaplan-Meier survival analysis was first used to calculate the probability of seizure-freedom in the overall group prior to any outcome predictor analysis. Comparison of continuous variables was examined by two-sample t test with equal variances and Mann-Whitney U test. Statistical analysis of categorical variable was performed using the chi-square test and Fisher's exact test as appropriate. A Logistic regression model was used to examine the association between favorable seizure outcomes and clinical variables. Statistical significance level was set at P <0.05.

RESULTS

Patient characteristics

In our surgical cohort, 99 patients fulfilled all criteria and were included in the analysis. Detailed demographic and patient information are listed in Table 1.

Table 1. Demographic data and clinical characteristics of patients

Characteristics	Data
Mean age at seizure onset (years)	15.17±11.06 (0–55)
Mean age at surgery (years)	28.35±12.00 (8–59)
Mead IPI age (years)	10.1±13.2 (0–55)
Mean follow-up time (months)	43.8±21.4 (12–72)
Gender (female/male) (*n* (%))	36 (36.4)/63 (63.6)
IPI (*n* (%))	
Febrile convulsion	2 (2.1)
Central nervous system infection	8 (8.1)
Head trauma	12 (12.1)
Cerebral vascular disease	1 (1.0)
Prenatal asphyxia	9 (9.1)
Others	2 (2.1)
Multiple types of epilepsy, *n* ≥2 (*n* (%))	47 (47.47)
SGTCS (*n* (%))	39 (39.39)
Aura (*n* (%))	32 (32.32)
Status (*n* (%))	8 (8.08)
Neurologic abnormal findings (*n* (%))	7 (7.07)
Familiy history (*n*)	0
Frequency of seizure (*n* (%))	
Daily	25 (25.25)
Weekly	49 (49.49)
Monthly	18 (18.18)
Yearly	2 (2.02)
Refractory (*n* (%))	64 (64.65)

SGTCS: secondary generalized tonic clonic seizures.

Postoperative mortality and morbidity

No postoperative mortality was found. Postoperative complications occurred in 13 patients. Seven of these had postoperative infections and four had postoperative epidural or subdural hematoma. Neurological deficits resolved before discharge in all patients except for two patients with long-term sequela including aphasia in one case and paralysis in another.

Overall seizure recurrence

Figure shows the Kaplan-Meier survival curve illustrating seizure recurrence, with 79 patients (79.8%) having experienced seizure recurrence at the time of the final follow-up. The chance of seizure freedom varied widely with post-operative time. The estimated chance of seizure freedom was 76.8% in the first month after surgery, and it fell abruptly to 51.5% within the first 3 months, then decreased at a somewhat slower rate between 3 and 24 months, and stabilized at around 20% percent within 2 to 6 years. The seizure-free rate was higher for temporal lobe resection than for extratemporal lobe surgery, but the difference did not reach statistical significance (*P* >0.05). Those patients reccurred after 6 months postoperatively tended to have a favorable seizure outcome compared with those with recurrence within the first 6 months (*P* <0.001). At the time of the final follow-up, evaluation using the ILAE criteria showed that there were 15 patients in Class 1a, 5 in Class 1, 5 in Class 2, 21 in Class 3, 22 in Class 4, 30 in Class 5 and 1 in Class 6. For the 79 patients experiencing seizure recurrence, there were 3 in Class 1, 2 in Class 2, 21 in Class 3, 22 in Class 4 and 30 in Class 5 in the last follow-up. Thus, despite seizure recurrence, 34% of patients were significantly improved at the time of the final follow-up compared with before surgery.

Univariate analysis

We classified all patients into favorable outcome and unfavorable outcome according to the ILAE classification. Classes 1a, 1 and 2 were regarded as favorable outcomes, and Classes 3–6 were regarded as unfavorable outcomes. At the last follow-up, 27 patients were classified into the favorable outcome group and 72 patients were classified into the unfavorable outcome group.

Seizure outcome in relation to clinical data

The influence of clinical variables on seizure outcome is shown in Table 2. There were no significant differences between individual subgroups of patients. No clinical factors were found to be significant predictors of postsurgical outcome.

Seizure outcome in relation to EEG features

We found that patients with highly localized ictal epileptiform activity and regional EEG patterns in the ECoG had better seizure outcomes compared with those with non-regional epileptiform abnormalities (*P* <0.05). However, there were no relationships between seizure outcome and background activity, focal slow wave or interictal EEG patterns.

Seizure outcome in relation to MRI features

Twenty patients had completely normal MRI findings and 79 had abnormal MRI results. Of the 79 patients with abnormal MRI findings, 57 had MRI-visible lesions concordant with abnormal EEG patterns, and the remaining 22 had MRI abnormalities that were not concordant with the EEG. We found significant differences between these three groups (*P* <0.05, Table 3). Patients with abnormal MRI findings concordant with EEG had more favorable seizure outcomes compared with those with abnormal MRI findings not concordant with EEG (*P* <0.05).

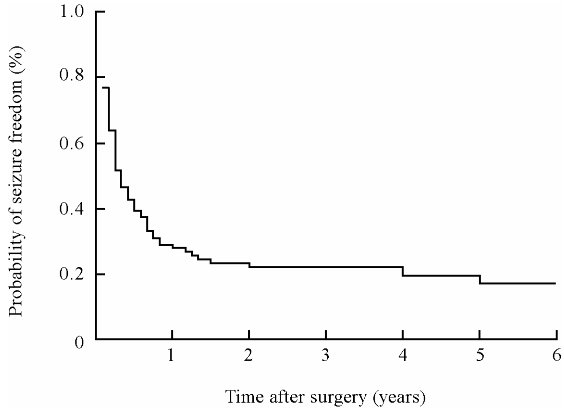

Figure. Kaplan-Meier plot illustrating chances of postoperative seizure freedom following resective surgery in overall cohort.

Table 2. Clinical data of the overall cohort, with comparison between the favorable and unfavorable outcome group

Variables	Favorable outcome (*n*=27)	Unfavorable outcome (*n*=72)	Statistic	*P* values
Age at onset (years)	13.89±1.53	15.65±1.42	0.06	0.950
Age at surgery (years)	25.52±1.49	29.42±1.55	0.90	0.368
Duration of disease (years)	11.63±1.57	13.86±1.10	1.09	0.276
Male (*n* (%))	45 (62.50)	18 (66.67)	0.14	0.701
Pre-operative auras present (*n* (%))	7 (25.93)	25 (34.72)	0.69	0.405
SGTCS present (*n* (%))	13 (48.15)	26 (36.11)	1.19	0.275
Multiple clinical seizure types, *n* ≥2 (*n* (%))	13 (48.15)	34 (47.22)	0.01	0.935
Status epilepticus present (*n* (%))	1 (3.70)	7 (9.72)	–	0.441
IPI present (*n* (%))	11 (40.74)	23 (31.94)	0.67	0.412
Neurological abnormal findings (*n* (%))	0	7 (9.72)	–	0.185
Refractory (*n* (%))	20 (74.07)	44 (61.11)	1.44	0.230
Daily seizure, daily (*n* (%))	5 (18.52)	20 (27.78)	0.89	0.345

P values are for χ^2 or Fisher's exact teats as appropriate.

Seizure outcome in relation to pathology

The pathological findings of the resected epileptic lesions are presented in Table 4. Fourteen patients had dual pathology, defined as HS plus another pathology such as mild focal cortical dysplasia or tumor. These included 12 patients with HS, one with cavernous hemangioma, and another with encephalomalacia. Different pathology had significant differernce compared with each other and the HS group had favorable outcomes compared with the group without HS, but there were no significant differences compared with the other groups.

Seizure outcome in relation to surgical variables

Analysis of the relationships between surgical variables and seizure outcome showed that patients with complete resections had superior outcomes compared with patients with incomplete resections (*P* <0.05). Patients with post-operative complications had unfavorable seizure outcomes (*P* <0.05). Other surgical variables did not

Table 3. Seizure outcomes in relation to MRI features (*n* (%))

MRI features	Favorable outcome	Unfavorable outcome	*P* values
Normal (*n*=20)	3 (11.1)	17 (23.6)	0.045*
Abnormal (*n*=79)			
Concordant with EEG (*n*=57)	21 (77.8)	36 (50.0)	0.011†
Inconcordant with EEG (*n*=22)	3 (11.1)	19 (26.4)	

*Seizure outcome of normal MRI group compared with concordant abnormal MRI group and inconcordant abnormal MRI group. †Seizure outcome of abnormal MRI concordant with EEG pattern compared with abnormal MRI inconcordant with EEG pattern.

Table 4. Seizure outcomes in relation to pathology (*n* (%))

Pathology	Favorable outcomes (*n*=27)	Unfavorable outcomes* (*n*=72)
Focal cortical dysplasia	6 (22.22)	31 (43.06)
Hippocampal sclerosis	10 (37.04)	3 (4.17)
Vascular malformation	1 (3.70)	2 (2.78)
Encephalmacia	1 (3.70)	11 (15.28)
Tumor	3 (11.11)	4 (5.56)
Dual pathology	3 (11.11)	9 (12.50)
Crypotogenic	2 (7.41)	11 (15.28)
FCD + other	1 (3.70)	1 (1.39)

*P <0.05, compared with favorable outcomes.

Table 5. Variables correlating with complete postoperative seizure-freedom after applying multivariate proportional hazard modeling

Variables	*OR*	SE	*Z* values	*P* values	95% *CI*
Recurred within 6 months	0.19	0.13	–2.40	0.016	0.05–0.74
Interictal regional EEG	0.52	0.38	–0.89	0.375	0.12–2.19
Ictal regional EEG	1.40	1.03	0.46	0.647	0.33–5.95
Subdural electrode	0.61	0.46	–0.65	0.517	0.14–2.68
ECoG	1.50	1.09	0.56	0.579	0.36–6.24
MRI concordant with EEG	6.33	4.78	2.44	0.015	1.44–27.81
Hippocampal sclerosis	18.06	12.86	4.07	0.000	4.48–72.88
Complete resection	0.60	0.69	–0.44	0.659	0.06–5.76

OR: odds ratio. SE: standard error.

influence outcome.

Multivariate analysis

Multivariate analysis showed that medial temporal lobe epilepsy with hippocampal sclerosis (P <0.01) and MRI with visible focal lesion concordant with EEG (P <0.05) were independent presurgical predictors of a favorable outcome (P <0.01). Seizure recurrence within 6 months was the only significant independent predictor associated with an unfavorable outcome (P <0.01, Table 5).

DISCUSSION

Prognosis analysis of resective surgery

In this study, we aimed to analyze long-term seizure outcomes after resective epilepsy surgery, according to the ILAE classification, and identify reliable prognostic factors for the selection of ideal patients. Kaplan-Meier survival analysis of the rates of seizure freedom showed that the probability of seizure freedom was 27.9% by the end of the first year and stabilized at 20% in the 2–6 years following surgery. Seizure outcome appeared to be better for temporal than for extratemporal lesionectomy, but this was not statistically significant. Previously reported rates of seizure freedom following resective surgery have varied. Stavem et al[11] reported that the two-year seizure-free rate after temporal lobe resection was only 44%, and Chaudhry et al[12] reported a rate of 63% after non-temporal lobe resection in 386 epilepsy subjects. Another study on the long-term prognosis (more than 5 years) after epilepsy surgery showed that only 16% of temporal lobectomy patients and 18% of patients with other surgery methods were completely seizure-free.[13] These wide differences may potentially be explained by heterogeneity of patient populations, evolution of surgical techniques between studies, and inconsistencies in the definition of a “favorable outcome”. Our finding of an abrupt drop in the seizure-free rate in the first three months followed by a relatively mild decline, is consistent with previous studies.[14,15]

Predictors of outcome in the resective surgery cohort

Consistent with previous reports,[16,17] we found no significant effect of sex, age at seizure onset, history of previous epilepsy surgery, seizure duration, seizure type, seizure frequency, auras, abnormal neurological signs, or the presence of IPI.

Preoperative interictal and ictal dynamic video EEG can greatly improve the detection of epileptic lesions. Our study showed that regional epileptiform patterns confirmed by ictal EEG and ECoG were predictors of a favorable surgery outcome. Yun et al[17] reported that a regional EEG pattern detected by ictal EEG recording was an independent predictive factor of seizure freedom. Others have found that an ictal EEG with regional EEG patterns was the only favorable outcome predictor for temporal lobectomy, even in patients with normal high-resolution MRI findings.[4] Our study found no association between post-operative outcome and abnormal EEG background, focal slow waves, or interictal regional EEG. Some studies have found that generalized interictal and ictal EEG pattern was predictive of a poor outcome.[18]

Our study showed that the outcome in patients with normal MRI presentation or with non-specific abnormalities was significant different from that in patients with MRI abnormalities concordant with EEG. After multivariate regression analysis, abnormal MRI with structural lesion was still an independent predictor of a favorable outcome, consistent with several other studies.[11,19,20] The constantly improving MRI technology can thus be used to locate epileptogenic lesions earlier and with better accuracy, and thereby improve the diagnosis of epilepsy and selection of candidates for surgery. This will avoid unnecessary disease prolongation and improve the efficacy of surgery.

In previous studies, focal cortical dysplasia (FCD) has been histopathologically proven in at least 20% of adult patients undergoing epilepsy surgery, and in almost 50%

of children.[19] Combining MRI and histopathology findings, our study showed that FCD constituted 52% of the lesions. This large proportion may be due to the fact that the age of seizure onset was relatively young in our cohort. The second most common pathology was HS (25.2%), with 13 subjects showing simple HS and 12 subjects showing dual pathology. We found that only simple HS predicts a favorable outcome (P <0.001), whereas tumors and vascular malformations did not significantly predict outcome. Other studies have found that at least 80% of tumor subjects and 70% of vascular malformation subjects obtained seizure freedom postoperatively.[21] Our opposite results may be related to the small number of patients.

The side of surgery (left/right), location (temporal/extratemporal), APO and reoperation all had no influence on surgery prognosis. Incomplete resection of the epileptogenic region predicted a poor outcome (P <0.05), similar to other studies,[22] but incomplete resection had no significant predictive effect after multivariate regression analysis. We concluded that this may be due to the distribution of pathologies in our cohort, with FCD in more than half of patients. Complete removal of FCD is often difficult because of the poor demarcation of the lesion, and dysplastic tissues tend to be more extensive than is apparent on MRI.

We also found that patients with seizure recurrence within the first 6 months showed poorer long-term prognosis than those with recurrence after 6 months. This finding was statistically significant in both univariate and multivariate analysis. Complications after surgery also indicated poor outcome in univariate analysis.

In conclusion, our study shows that hippocampal sclerosis and abnormal MRI with epileptic lesions are strongly associated with a favorable outcome after surgery, whereas seizure recurrence within 6 months of surgery was associated with a poor outcome. Successful surgery can lead to seizure freedom in previously intractable patients, and dramatically improve the quality of life in many patients. Thus, it is important to gain better knowledge of prognostic factors, to allow the selection of ideal candidates for epilepsy surgery.

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Correction

In the original article entitled *Changes of glucocorticoid receptor mRNA expression in basolateral amygdale-kindled rats* published in the September 5 issue 2011 (Chin Med J 2011; 124 (17): 2622-2627), the address of the authors BAO Guan-shui, CHEN Xu-qin, HUA Yin, WANG Zhe-dong and LIU Zhen-guo should have been: Department of Neurology, Xin Hua Hospital Affiliated to Shanghai Jiao Tong University School of Medicine, Shanghai 200092, China (Bao GS and Liu ZG); Department of Neurology, Affiliated Children's Hospital of Soochow University, Suzhou, Jiangsu 215003, China (Chen XQ and Wang ZD); Department of Child Neurology, Wuxi People's Hospital, Wuxi, Jiangsu 214001, China (Hua Y).